GLOBAL NEEDS · ACADEMIC PRESENTATION

Global Needs: Deforestation

Understanding the problem, its human impact, and practical ways forward

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COLLEGE PRESENTATION



A PROBLEM THAT CONNECTS CLIMATE, BIODIVERSITY, LIVELIHOODS, AND GOVERNANCE.

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01 / WHY IT MATTERS

Deforestation is a global problem with local consequences

Deforestation removes natural protection against heat, floods, soil erosion, and biodiversity loss. One land-use decision can affect climate, wildlife, and communities at the same time.

GLOBAL SCALE

4.06B

About 4.06 billion hectares of forest cover nearly one-third of the world's land area.

SOURCE: FAO

LOCAL PRESSURE

Forest-dependent people face pressure on food, fuel, water, health, and livelihoods.

Key question: how can development meet human needs without destroying the systems that support life?



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Introduction: what deforestation changes

Forests are living infrastructure, not empty land.



Land use

Deforestation is the permanent removal of trees and forest cover to convert land for other uses.



Drivers

Agricultural expansion, cattle grazing, logging, mining, roads, fires, and settlement growth.



Ecology

Forest loss breaks habitats into smaller areas, weakens ecosystem services, and can increase exposure to extreme weather.



People

Some forest products are essential to households, so solutions must protect both nature and people’s needs.

THE SYSTEM

Climate · water · soil · biodiversity · communities are connected through the forest.

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Survey and 2020 evidence: the pressure is measurable

The rate of net loss slowed, but forest area continued to decline.

GLOBAL FOREST AREA TREND, 1990–2020



SOURCE: FAO GLOBAL FOREST RESOURCES ASSESSMENT 2020

WHAT THE ASSESSMENT COVERED

236 countries and territories

FAO examined more than 60 forest variables across 1990–2020.

TOTAL CHANGE

178M hectares

Global forest area declined by about 178 million hectares between 1990 and 2020, roughly the size of Libya.

SOURCE: FAO

LOCAL SURVEY QUESTIONS

Where do households get fuel and timber? Have rainfall, heat, or water availability changed? Which forest rules are difficult to follow?

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Solutions should protect forests and people together

Protection lasts when it supports food, energy, rights, and livelihoods.



AGROFORESTRY

Support agroforestry and climate-smart farming so food production does not depend on clearing new forest.



LOCAL RIGHTS

Strengthen community forest management, secure local rights, and reward verified protection.



RESTORATION + ENERGY

Restore native ecosystems with local species; measure survival, water, soil, and livelihood outcomes. Reduce household pressure through efficient cooking, clean energy, and affordable alternatives to illegal fuelwood.

TRACE TIMBER, PALM OIL, SOY, CATTLE, AND OTHER HIGH-RISK SUPPLY CHAINS; MEASURE SURVIVAL, WATER, SOIL, AND LIVELIHOO

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Existing systems: useful tools, incomplete protection

The gap is not a lack of tools. It is the missing connection between detection, action, and benefit.

CURRENT PROTECTION LOOP



Protected areas

Conserve important habitats.

Monitoring

Detect and compare forest change.

Local action

Communities need voice, reliable livelihoods, and long-term support.

Accountability

Laws, certification, checks, and restoration create standards.

INCOMPLETE PROTECTION

- Boundaries cannot stop pressure outside protected areas.
- Detection does not automatically trigger a response.
- Enforcement and transparent data remain uneven.
- Funding and local participation are often short-term.

Assessment: connect monitoring to rapid action and community benefit, rather than treating them as separate projects.

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06 / PROPOSED SOLUTION

Proposed solution: a community-led forest protection loop

Use evidence to act early, then let local results improve the next decision.

01

Map risk

Combine satellite data with local knowledge.

02

Co-design

Plan protection and livelihoods with communities, farmers, and local government.

03

Act early

Use patrols, fire prevention, restoration, and sustainable production support.

04

Measure

Publish forest cover, biodiversity, water, income, and participation measures.

05

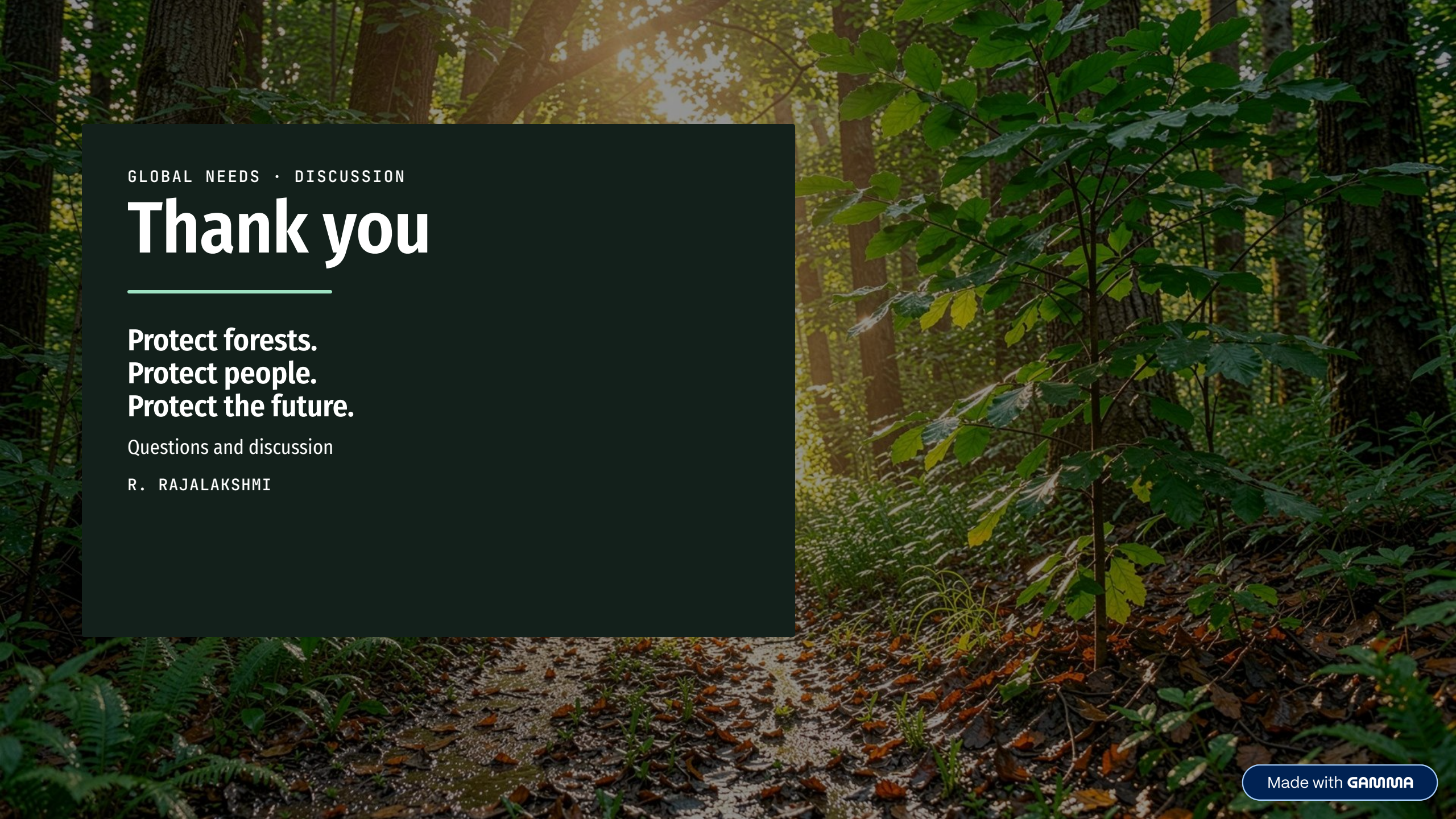
Review

Assess results every season and redirect resources to what works.

CONCLUSION

Evidence + local ownership + responsible consumption + restoration + accountable governance.

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GLOBAL NEEDS · DISCUSSION

Thank you

Protect forests.
Protect people.
Protect the future.

Questions and discussion

R. RAJALAKSHMI

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